ASTR 1020 Lab

L6: Hubble's Law

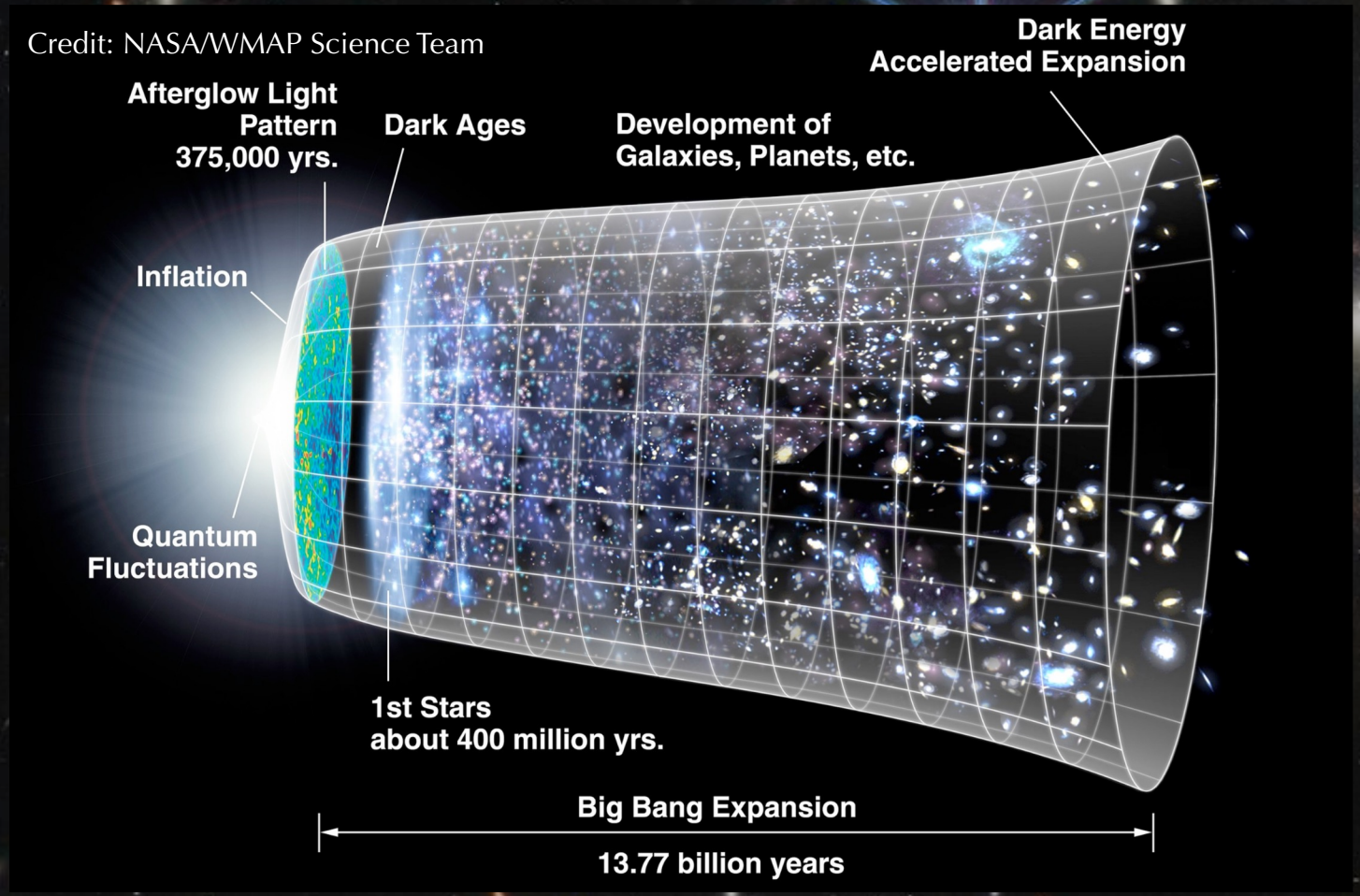


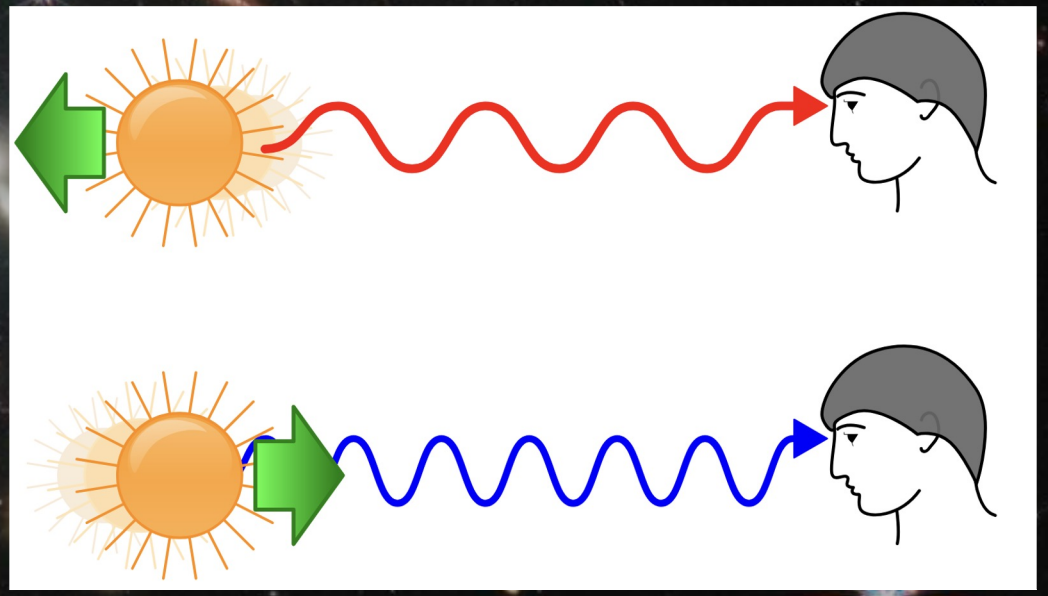
Credit: NASA, ESA, CSA, STScI

Things You Need To Know For Lab 6

- The expanding universe
- Redshifts and blueshifts (**Doppler** shifts)
- Recessional velocities
- Hubble's Law

Expanding Universe & Doppler Shifts





Credit: Aleš Tošovský

Redshift & Recessional Velocities

$$z = \frac{\lambda_{observed} - \lambda_{rest}}{\lambda_{rest}}$$

$$v = z \times c$$

Redshift

Recessional velocity

Speed of light

Hubble's Law

- Hubble constant, H_0, measures expansion rate of the Universe
- Recessional Velocity = $H_0 \times$ Distance
- Get value of H_0 from trendline equation!
- Use Excel file on iCollge page